HxGN OnCall® Dispatch

Upgrade Guide

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About this Release

This document describes the HxGN OnCall Dispatch 10.0.2309.4 release. The information in this document is current as of the product release.

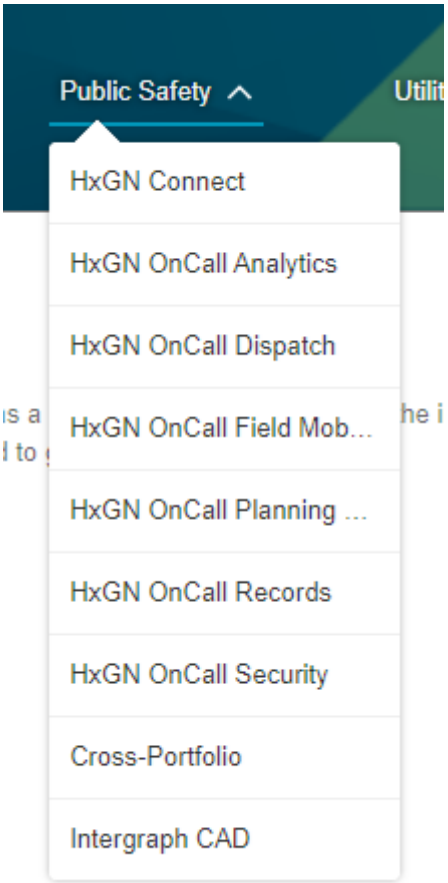
This document is only an introduction and does not provide all the details about the product capabilities. See the Help and other documents provided with the HxGN OnCall Dispatch products for more information.

What's New

To read about new features, updated features, and system requirements, use the **Support** page. For online and phone support, see *Technical Support* (on page 53).

1. Go to the *Support page* *https://hexagon.com/support-success/safety-infrastructure-geospatial*.
2. Click **Log in**.
3. Type your username and password, and click **Log in**.

4. From the **Products** list, select your product portfolio.



The list of products for the selected portfolio displays.

5. Click a product link to open its support page.

6. When the product page opens, click the appropriate links to view product announcements or technical documentation.
   - Click **Release Notes** to review new or enhanced product features.
   - Click **Issues Resolved** to review defects that have been fixed.
   - Click **Supported Environments** to review system requirements.

- Click **Open-Source Software Notices** in the **Product Information** box, if available, to review information on open-source software in the product.

Some minor releases might not provide Release Notes or Issues Resolved.

Upgrade Your HxGN OnCall Dispatch System

HxGN OnCall® Dispatch periodically releases cumulative updates. If you already have an earlier version of HxGN OnCall Dispatch installed, follow the instructions below to upgrade your system.

In OnCall Dispatch systems running version 2303 or later, ISM is used to run the websites instead of IIS. It is necessary to stop 2209 websites in IIS, but the 2303 websites are started in ISM. Add-on products may or may not still use IIS. Hexagon recommends reviewing the documentation for each add-on for more information.

Stop HxGN OnCall Dispatch Processes

To stop processes on your HxGN OnCall Dispatch system:

1. Open a PowerShell window using the **Run as administrator** option on each server.
2. Stop the ISM service on each server by running the following PowerShell command.

   ```
   Stop-Service ism
   ```

3. Close any administration tools on each server. Examples of administration tools include ISM Administrator, CAD Database Utility, CAD Query Tool, and CAD Bulk Loader.

   If you have any add-on products, you need to stop websites for them, as well.

Upgrade Administration Tools

HxGN OnCall Dispatch administration tools only need to be upgraded on servers where they are already installed. However, the following instructions can be used to install the administration tools on additional servers, as well.

To upgrade the administration tools:

1. Open a PowerShell window using the **Run as administrator** option.
2. From the PowerShell window, define the **OnCallDeliveryRoot** environment variable.

   The **OnCallDeliveryRoot** variable is wherever your delivery media is installed.

Example: `$env:OnCallDeliveryRoot = "C:\delivery\IPS1122 - HxGN OnCall Dispatch - Advantage\Advantage"`

3. Update the **Admin Tools** using the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
-AdminTools
```

This also installs the **Admin Tools** user guides (PDFs).

Upgrade Your Live and Archive Databases

The following procedures feature command examples for upgrading your database using PowerShell. However, if you are using Windows Authentication and a service account for upgrading your database, the command examples would instead look like the following:

```
-DbConfigName <DatabaseConfigName> -DbOwner "\"
-ArchiveDbConfigName <ArchiveDBConfigName> -ArchiveDbOwner "\"
```

Password parameters, such as ArchiveDbOwnerPassword, are not necessary to include when Windows Authentication is being used.

To upgrade your live and archive databases:

1. Make a backup of your live and archive databases before upgrading.
2. Open a PowerShell window on one of the application servers where the administration tools have been installed.
3. From the PowerShell window, define the following environment variables using the *SetUpgradeVariables.ps1* file:
    - **OnCallSystemRoot** - This is where <aProductName> is or can be installed.
    - **OnCallDeliveryRoot** - This is wherever your delivery media is installed.
    - **OldVersion** - This is the current <aProductName> version.
    - **NewVersion** - This is the new <aProductName> version.
    - **DbConfigName** - This is the database configuration name for the live system.
    - **DbOwner** - This is the database owner for the live system.
    - **DbOwnerPassword** - This is the database owner password for the live system.

- **ArchiveDbConfigName** - This is the database configuration name for the archive system.

  If there is no archive database, leave this variable empty.

- **ArchiveDbOwner** - This is the database owner for the archive system.

  If there is no archive database, leave this variable empty.

- **ArchiveDbOwnerPassword** - This is the database owner password for the archive system.

  If there is no archive database, leave this variable empty.

- **StringLocales** - This is a list of one or more string locales, separated by semi-colons.

  An example of a full environment with defined variables looks like the following:

```
& "C:\delivery\IPS1122 - HxGN OnCall Dispatch -
Advantage\Advantage\SetUpgradeVariables.ps1"
-OnCallSystemRoot "C:\Program Files\Intergraph"
-OnCallDeliveryRoot "C:\delivery\IPS1122 - HxGN OnCall
Dispatch - Advantage\Advantage" -OldVersion 10230300
-NewVersion 10230900 -DbConfigName live_db_config
-DbOwner live_db_config_owner -DbOwnerPassword
"password" -ArchiveDbConfigName archive_db_config
-ArchiveDbOwner archive_db_config_owner
-ArchiveDbOwnerPassword "password" -StringLocales
"en-us;fr-fr;de-de"
```

Upgrade Your Live and Archive Databases

Ensure that you correctly follow the instructions necessary for your upgrade. If you are changing major versions (for example, upgrading from 2303 to 2309), follow the major release instructions. Steps are labeled based on the upgrade type where applicable. When not labeled, the step applies to all release upgrade types.

4. **(Major Release Upgrades Only)** To upgrade your live database schema, make the schema changes to the *live* database by running the following PowerShell command:

```
& "$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole"
-ExecuteUpgrade -BeginVersion "$env:OldVersion" -DbConfigName
"$env:DbConfigName" -User "$env:DbOwner" -Password
```

```
"$env:DbOwnerPassword" -EndVersion "$env:NewVersion"
-Interactive
```

- You are prompted for values during the upgrade.
- This process can take some time to complete.

5. **(Minor Release and Patch Upgrades Only)** To make non-breaking changes to your live database schema, run the following PowerShell command:

```
& "$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole"
-ExecuteMisc -SchemaFile
"$env:OnCallSystemRoot\sql\upgrade\$($env:NewVersion)fixes.x
ml" -DbConfigName "$env:DbConfigName" -User "$env:DbOwner"
-Password "$env:DbOwnerPassword" -Interactive
```

You may be prompted for values during the execution of the file.

6. Verify schema changes are consistent with the definition file for the live database by running the following PowerShell command:

```
& "$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole"
-ExecuteDefinition -SchemaFile
"$env:OnCallSystemRoot\sql\db_schema_both.xml" -DbConfigName
"$env:DbConfigName" -User "$env:DbOwner" -Password
"$env:DbOwnerPassword" -Interactive
```

- You are prompted for values during the definition.
- This process can take some time to complete.

7. **(Major Release Upgrades Only)** Make the schema changes to the *archive* database by running the following PowerShell command:

```
& "$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole"
-ExecuteUpgrade -BeginVersion "$env:OldVersion" -DbConfigName
"$env:ArchiveDbConfigName" -User "$env:ArchiveDbOwner"
-Password "$env:ArchiveDbOwnerPassword" -EndVersion
"$env:NewVersion" -Interactive
```

- You are prompted for values during the upgrade.
- This process can take some time to complete.

8. **(Minor Release and Patch Upgrades Only)** To make non-breaking changes to your *archive* database schema, run the following PowerShell command:

```
& "$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole"
-ExecuteMisc -SchemaFile
"$env:OnCallSystemRoot\sql\upgrade\$($env:NewVersion)fixes.x
ml" -DbConfigName "$env:ArchiveDbConfigName" -User
"$env:ArchiveDbOwner" -Password "$env:ArchiveDbOwnerPassword"
-Interactive
```

   You may be prompted for values during the execution of the file.

9. Verify schema changes are consistent with the definition file for the archive database by running the following PowerShell command:

```
& "$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole"
-ExecuteDefinition -SchemaFile
"$env:OnCallSystemRoot\sql\db_schema_both.xml" -DbConfigName
"$env:ArchiveDbConfigName" -User "$env:ArchiveDbOwner"
-Password "$env:ArchiveDbOwnerPassword" -Interactive
```

   - You are prompted for values during the upgrade.
   - This process can take some time to complete.

10. Execute the *setup_archive_database.xml* file on the *archive* database by running the following PowerShell command:

```
& "$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole"
-ExecuteMisc -SchemaFile
"$env:OnCallSystemRoot\sql\setup_archive_database.xml"
-DbConfigName "$env:ArchiveDbConfigName" -User
"$env:ArchiveDbOwner" -Password "$env:ArchiveDbOwnerPassword"
```

11. To upgrade your parameters, command lines, localized strings, and other configuration data, run the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\ConfigurationDataUpgrade.ps1"
-OnCallSystemRoot  "$env:OnCallSystemRoot" -dbOwner
"$env:DbOwner" -dbOwnerPassword "$env:DbOwnerPassword"
-StringLocales "$env:StringLocales"
```

12. Update feature flags by running the following OcUtil command in a PowerShell window:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil" FeatureFlags
-f
"$env:OnCallSystemRoot\OnCall\OcUtil\ConfigurationData\Featu
reFlags\feature_flags.config.json " -v -d
```

13. **(Major Release Upgrades Only)** Cleanup old schema objects and old parameters and strings from databases by running the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\DatabaseUpgradeCleanup"
-OnCallSystemRoot  "$env:OnCallSystemRoot" -LiveDbConfig
"$env:DbConfigName" -LiveDbOwner  "$env:DbOwner"
-LiveDbPassword "$env:DbOwnerPassword" -ArchiveDbConfig
"$env:ArchiveDbConfigName" -ArchiveDbOwner
"$env:ArchiveDbOwner" -ArchiveDbPassword
"$env:ArchiveDbOwnerPassword"
```

Upgrade Your Database Management System

To upgrade Oracle or SQL Server:

1. If necessary, perform a rolling upgrade for the Database Management System (DBMS) on the *live* database servers.
2. If necessary, perform a rolling upgrade for the DBMS on the *archive* database servers.

Upgrade Your Application Servers

Upgrade your application servers one at a time, starting with the secondary servers. For each server, perform the following steps.

To upgrade application servers on your HxGN OnCall Dispatch system:

1. Back up any changed or custom configuration files. These may include the following:
   - ISM configuration files located in *C:\ProgramData\Intergraph\Cad\ISM*
   - The *appsettings.json* files located in application folders
   - The *log4net.config* files located in application folders

     The default install location for HxGN OnCall Dispatch applications is *C:\Program Files\Intergraph*.
2. Open a PowerShell window using the **Run as administrator** option.

3. From the PowerShell windows, define the following environment variables:
   - **OnCallSystemRoot** - This is where <aProductName> is or can be installed. This is typically *C:\Program Files\Intergraph*, for example, `$env:OnCallSystemRoot = "C:\Program Files\Intergraph"`
   - **OnCallDeliveryRoot** - This is wherever your delivery media is installed. For example, `$env:OnCallDeliveryRoot = "C:\delivery\IPS1122 - HxGN OnCall Dispatch - Advantage\Advantage"`
4. Update the services by running the following PowerShell command:

   ```
   & "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
   -Services
   ```

   This process can take some time.
5. Update the interfaces by running the following PowerShell command:

   ```
   & "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
   -Interfaces
   ```

   Product folders for interfaces must be placed one folder location above your **OnCallDeliveryRoot** in order for the PowerShell script to run. For example, if you set the **OnCallDeliveryRoot** to "*C:\delivery\IPS1122 - HxGN OnCall Dispatch - Advantage/Advantage*," then the interface product folders need to be in "*C:\delivery\IPS1122 - HxGN OnCall Dispatch - Advantage*."
6. Update ISM configuration files. This step overwrites any customized files.
   a. On a primary server, run the following PowerShell command:

      ```
      & "$env:OnCallDeliveryRoot\OrganizeIsmConfigs.ps1"
      -InstallDirectory $env:OnCallSystemRoot
      ```

   b. On a secondary server, run the following PowerShell command:

      ```
      & "$env:OnCallDeliveryRoot\OrganizeIsmConfigs.ps1"
      -InstallDirectory $env:OnCallSystemRoot -Secondary
      ```
7. Merge any ISM configuration file customizations back into the ISM configuration files.
   a. Compare the new files against the backups that were made earlier.
   b. Update the new files, as necessary.

Upgrade Your Web Servers

For each server, perform the following steps.

To upgrade web servers on your HxGN OnCall Dispatch system:

1. Back up any changed or custom configuration files. These may include the following:
   - ISM configuration files in *C:\ProgramData\Intergraph\Cad\ISM*
   - The *appsettings.json* files located in application folders
   - The *log4net.config* files located in application folders

     The default install location for HxGN OnCall Dispatch applications is *C:\Program Files\Intergraph*.
2. Open a PowerShell window using the **Run as administrator** option.
3. From the PowerShell windows, define the following environment variables:
   - **OnCallSystemRoot** - This is where <aProductName> is or can be installed. This is typically *C:\Program Files\Intergraph*, for example, `$env:OnCallSystemRoot = "C:\Program Files\Intergraph"`.
   - **OnCallDeliveryRoot** - This is wherever your delivery media is installed. For example, `$env:OnCallDeliveryRoot = "C:\delivery\IPS1122 - HxGN OnCall Dispatch - Advantage\Advantage"`.
4. Update the web applications that were taken down by running the following PowerShell command:

   ```
   & "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
   -WebApps
   ```
5. Update ISM configuration files. This step overwrites any customized files.
   a. On a primary server, run the following PowerShell command:

      ```
      & "$env:OnCallDeliveryRoot\OrganizeIsmConfigs.ps1"
      -InstallDirectory $env:OnCallSystemRoot
      ```

   b. On a secondary server, run the following PowerShell command:

      ```
      & "$env:OnCallDeliveryRoot\OrganizeIsmConfigs.ps1"
      -InstallDirectory $env:OnCallSystemRoot -Secondary
      ```
6. Merge any ISM configuration file customizations back into the ISM configuration files.

a. Compare the new files against the backups that were made earlier.

b. Update the new files, as necessary.

The following is an optional step for sites using OnCall.IWAServer for Windows Authentication.

- If upgrading a system that has OnCall.IWAServer installed and configured, open an Administrator command prompt and run the delivered *install_oncall.iwaserver.bat* file.
- If doing a fresh installation of OnCall.IWAServer, see the "Configure Windows Authentication" topic in the *HxGN OnCall Dispatch Installation and Configuration Guide.*

If service accounts are used for the system, the service account must have write permission to *C:\Program Files\Intergraph\OnCall.Website\StaticFiles*. Icons on the map do not display if this is not done.

Configure Your Web Servers

Remove Old Application Pools

Since OnCall Dispatch systems on version 2303 or later use ISM to run the websites, their application pools may be removed. For each web server, perform the following steps.

To remove old application pools:

1. Open a **PowerShell** window using the **Run as administrator** option.
2. From the PowerShell window, define the **OnCallDeliveryRoot** environment variable. This is wherever your delivery media is installed. For example, `$env:OnCallDeliveryRoot = "C:\delivery\IPS1122 - HxGN OnCall Dispatch - Advantage\Advantage"`.
3. Clean up old application pools by running the following command:

```
& "$env:OnCallDeliveryRoot\CleanupAppPools.ps1"
```

Response Compression

The HxGN OnCall Dispatch Website compresses static JavaScript and CSS files. In general, file compression can help reduce network traffic and improve performance.

However, if you need to disable compression, set the **UseCompression** setting to **false** (`UseCompression=false`) in the corresponding service's command line in ISM.

Dynamic responses from OnCall.WebApi are compressed by default. You can disable response compression using the **UseCompression** setting (`UseCompression=false`) in that service's command line.

HTTPS Configuration

The following applications require using HTTPS for secure access:

- Proxy
- Identity
- RestfulCad (RestAPI)
- CadNotificationService
- CopyEnvironmentModule
- Deccan API
- OnCall OcUtilServiceModule
- OnCall.UnitEventModule

These applications can be configured using the global *appsettings.json* file and Certificate Manager or by modifying their individual JSON files. The global *appsettings.json* file is included in the cadglobalconfig chocolatey package (version 0.0.3 or later), which is installed alongside the CAD Database Utility. The file must be placed in the *C:\ProgramData\Intergraph\Cad* folder on any server running the applications listed above.

To configure HTTPS globally for the HxGN OnCall Dispatch applications:

1. Verify your certificate is located in the Trusted Root Certificates folder.
2. Navigate to */ProgramData/Intergraph/Cad*.
3. Using a text editor, open the *appsettings.json* file and locate the `Kestrel` section:

```
"Kestrel": {
   "Certificates": {
      "Default": {
     "Subject": "*.ingrnet.com",
     "Store": "root",
```

```
        "Location": "LocalMachine",
        "AllowInvalid": false
        }
      }
    }
```

4. Enter parameter values that match your certificate.
   - **Subject** – The subject of the certificate to use. This value can be found in the properties of a certificate by using Certificate Manager.
   - **Store**
     - `root` - the Trusted Root Certificates folder.
     - `my` - your personal certificates folder.
   - **Location**
     - `LocalMachine` - the system certificate store
     - `CurrentUser` - the certificate store of the user
   - **AllowInvalid** - Hexagon recommends setting this value to false for production environments. If set to true, invalid certificates are allowed; this value would typically be used when testing self-signed certificates.
5. Ensure the certificate is installed in your local trust store. Hexagon does not provide this certificate. The certificate used to support HTTPS may be used, but it must contain the private key and the public key in an RSA format. If it is not installed, install the certificate using the Certificate Import Wizard and use your local computer as the default location.

Other HTTPS Methods

If Certificate Manager cannot be used, the following alternate methods are supported:

Use a PFX Certificate File

1. In the installation folder for OnCall Dispatch, locate the *appsettings.json* file.
2. Open the JSON file with an editor, and locate the Kestrel section.
3. Replace the default Kestrel section with the following information:

```
"Kestrel": {
```

```
        "Certificates": {
          "Default": {
            "Path": "C:\\certificates\certificate.pfx",
            "Password": "<password>"
          }
        }
      }
```

4. Enter parameter values that match your certificate.
   - **Path** – the path to the PFX file
   - **Password** – the password to use the PFXfile

Use A PEM/CRT File

1. In the installation folder for OnCall Dispatch, locate the *appsettings.json* file.
2. Open the JSON file with an editor, and locate the Kestrel section.
3. Replace the default Kestrel section with the following information:

```
"Kestrel": {
    "Certificates": {
      "Default": {
          "Path": "<path to .pem/.crt file>",
          "KeyPath": "<path to .key file>",
          "Password": "$CREDENTIAL_PLACEHOLDER$"
      }
    }
  }
```

Enter parameter values that match your certificate.

- **Path** – the path to the PEM/CRT file
- **Keypath** – the path to the related KEY file
- **Password** – the password that was used to encrypt the KEY and CRT files

Changing Log Level for OnCall.RealTime.WebApi

To change the log level for OnCall.RealTime.WebApi:

1. Select the *OnCall.RealTime.WebApi.log4net.config* file found in the *\Program Files\Intergraph\OnCall.RealTime.WebApi* folder.
2. Change the *<root><level value="WARN">* and *<logger name="NetMessage"><level value="WARN">* values as needed.

Changing Log Level for OnCall.Identity

To change the log level for OnCall.Identity:

1. Stop **OnCall.Identity** if the service is running.
2. Navigate to the installed folder at *C:\Program Files\Intergraph\OnCall.identity*.
3. Open *OnCall.IdentityServer.dll.config* using a text editor.
4. Locate the *<add key="defaultDebugLevel" value="1" />* in the code.
5. Change the value to increase or decrease the amount of logging produced.

   Valid values range from 0 to 5. Zero being the least amount of logging, and five the most amount of logging.
6. Save the change.
7. Restart **OnCall.Identity**.

   The newly defined value log output is now used.

Configure Load Balancer Health Check Endpoints

Configure health check endpoints in your load balancer to confirm the availability of your web servers. Hexagon recommends using the **OnCall.Website** endpoint; however, other endpoints are available for testing purposes.

The following table lists the web applications and their endpoints.

Web Application	Endpoint
CadNotificationService	*/cadnotificationservice/api/healthcheck*
OnCall.Identity	*/oncall.identity/healthcheck*

OnCall.RealTime.WebApi	*/oncall.realtime.webapi/healthcheck*
OnCall.WebApi	*/oncall.webapi/healthcheck*
OnCall.Website	*/oncall/healthcheck*
RestfulCad (RestAPI)	*/restfulcad/api/healthcheck*

Reserve Ports for HxGN OnCall Dispatch Services

Many of the HxGN OnCall Dispatch services expose a web API. The HxGN OnCall Dispatch WebApi communicates with HxGN OnCall Dispatch services using HTTP requests and RabbitMQ messages. You may need to reserve the ports that the services are using for their web API. The *ReserveURLNamespace.bat* script runs the `netsh http add urlacl` command, which reserves the ports and URLs that the services need.

The *ReserveURLNamespace.bat* file is also delivered with HxGN OnCall Dispatch. The *ReserveURLNamespace.bat* file can be used as needed to add (define), show, or delete port reservations. The BAT file calls *ReserveURLNamespace.ps1*. For more information and examples, run the following command from a PowerShell window:

```
get-help .\ReserveURLNamespace.ps1
```

To reserve ports for the HxGN OnCall Dispatch services:

1. Open File Explorer on the computer hosting the HxGN OnCall Dispatch services.
2. Navigate to the *C:\Program Files\Intergraph\OnCall\Core\ConfigurationData\Scripts* folder.

   The following ports are reserved for these services. If needed, update the port numbers on your system by changing the values in your ISM configuration file and changing the port numbers in the *ReserveURLNamespace.bat* before running the script.

Service	Port
RestAPI	8080
CADNotificationService	8081
Deccan API	8088

Service	Port
Calltaker	9001
ContactManagerModule	9002
DialogsModule	9003
CadBulkLoaderService	9004
Alert Module	9005
DeviceNotificationModule	9006
Recommend Unit Service	9007
Tracker	9008
RoleAdminModule (AdminCommands)	9010
ConfigServer	9011
InquiryModule	9014
OnCall StringServerModule	9015
LogonModule	9016
ProduceWorkModule	9017
GeoFence Service	9018
Loi Service	9019
Informer	9020
FireStationPrintingModule	9021
FireStationAlertingModule	9022
DashboardModule	9023
DataServiceModule	9024
EventCommandsModule	9025

Service	Port
GeoModule	9026
GeoFenceModule	9027
UnitCommandsModule	9028
TimersModule	9029
OnCall OcUtilServiceModule	9030
AniAliModule	9031
Copy Environment	9032
SopServiceModule	9033
BroadcastModule	9034
AuditCommandModule	9035
AdUpdateModule	9036
DefinitionsCacheModule	9037
InformerProviderModuleCommSys	9038
InformerProviderModuleOnCallRecords	9039
InformerProviderModuleRabbitMQ	9040
InformerProviderModuleODBC	9041
FacilityCommandsModule	9042
CommunicationsModule	9043
FirestationPrintingProviderModuleRabbitMQ	9044
FirestationAlertingProviderModuleRabbitMQ	9045
DbArchive Service	9046

Service	Port
CADLink	9047
FIRELink	9048
RMSLink	9049
PushToTalkModule	9050
Outlook Calendar Service	9051
OnCall Identity	44318
OnCall WebApi	44321
OnCall RealTime WebApi	44322
OnCall Website	44323
OnCall Proxy	44343

3. Right-click *ReserveURLNamespace.bat* and select **Run as administrator**.

   You may need to restart the HxGN OnCall Dispatch services after making this change.

4. If the website still cannot connect to a BaseApp's web API, make sure you have configured the firewall on the servers to allow access.

   - You need to expose port 44343 on your server firewall. Next, you need to change the load balancer to access the website on port 44343.
   - If this is a test environment without a load balancer, the **ONCALL_BASE_URL** value in the *cad-global.json.config* file needs to be updated to include port 44343. For example, the URL would look like this: *https://servername.domain.com:44343/oncall*.

Configure the Firewalls

Configure the firewalls on your database servers, application servers, and web servers to allow access.

See the following topics for more information:

- "*Configure Firewalls* (on page 47)" for information about allowing traffic from Hexagon applications
- "Firewall Settings" in the *Oracle 19c Server Installation and Configuration Guide*
- "Configuring the Firewall to Allow SQL Server Traffic" in the *SQL Server Installation and Configuration Guide*

Start HxGN OnCall Dispatch Processes

To start processes on your HxGN OnCall Dispatch system:

1. Start the ISM service on the application servers by running the following PowerShell command for each server:

   ```
   Start-Service ism
   ```

   a. Use ISM Administrator to monitor the status of your HxGN OnCall Dispatch services.

   b. Wait until all of the services are running before proceeding.

   c. You may experience a heavy CPU load during this process. Run **Task Manager** to verify the CPU usage and wait until it is back to a normal level.

2. If you have any add-on products, you may need to start websites for them, as well.

Zero Downtime Upgrade

The following procedure demonstrates how to perform an upgrade for HxGN OnCall Dispatch with zero downtime.

Ensure that you correctly follow the instructions necessary for your upgrade. If you are changing major versions (for example, upgrading from 2303 to 2309), follow the major release instructions. Steps are labeled based on the upgrade type where applicable. When not labeled, the step applies to all release upgrade types.

This process requires an upgraded test system to be used to assist with performing a zero downtime upgrade on a production system. Half of the production system is upgraded while the remaining half continues running. After the upgraded half is running again, the remaining half of the production system can be upgraded.

- Do not make any administrative changes to the production system during the upgrade process. Wait until the upgrade is complete before changing items such as parameters, strings, and configuration files.
- Once the zero downtime upgrade on the live system has started, it is not possible to rollback the changes without data loss and down time. It is vitally important to back up the databases and configurations files before starting the upgrade. This provides a means of recovery in case data is inadvertently corrupted or erased.
- Zero downtime upgrades require direct access to the load balancer in order to configure health check endpoints and disable specific servers during the upgrade.

Upgrade Your Test System

To upgrade a test system:

1. Follow the regular upgrade instructions on the test system until you get to Step 7 in the "*Upgrade Your Live and Archive Databases* (on page 10)" topic.
2. Open a PowerShell window using the **Run as administrator** option on one of the test system servers that as **Admin Tools** installed.
3. From the PowerShell window, define the following environment variables using the *SetTestSystemVariables.ps1* file:
    - **OnCallSystemRoot** - This is where <aProductName> is or can be installed.
    - **OnCallDeliveryRoot** - This is wherever your delivery media is installed.
    - **OldVersion** - This is the current <aProductName> version.
    - **NewVersion** - This is the new <aProductName> version.
    - **DbOwner** - This is the database owner for the live system.
    - **DbOwnerPassword** - This is the database owner password for the live system..
    - **CustomParameters** - This is the location of the parameter overrides that were generated on the test system.

      If there are no parameter overrides, leave this variable empty.
    - **CustomCommandLines** - This is the location of the command line overrides that were generated on the test system.

      If there are no command line overrides, leave this variable empty.

- **CustomStrings** - This is the location of the string overrides and localizations that were generated on the test system.

  If there are no string overrides or localizations, leave this variable empty.

- **CustomDialogs** - This is the location of the custom dialog boxes that were generated on the test system.

  If there are no custom dialog boxes, leave this variable empty.

  An example of a full environment with defined variables looks like the following:

```
& "C:\delivery\IPS1122 - HxGN OnCall Dispatch -
Advantage\Advantage\SetTestSystemVariables.ps1"
-OnCallSystemRoot "C:\Program Files\Intergraph"
-OnCallDeliveryRoot "C:\delivery\IPS1122 - HxGN OnCall
Dispatch - Advantage\Advantage" -OldVersion 10230300
-NewVersion 10230900 -DbOwner test_db_config_owner
-DbOwnerPassword "password" -CustomParameters
"c:\parameters" -CustomCommandLines "c:\commandlines"
-CustomStrings "c:\strings" -CustomDialogs "c:\dialogs"
```

4. **(Major Release Upgrades Only)** Start upgrade mode on the test system by running the following PowerShell command:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil"
ZeroDowntimeUpgrade --start --oldVersion "$env:OldVersion"
--newVersion "$env:NewVersion"
```

5. Resume the regular upgrade at Step 7 in the "*Upgrade Your Live and Archive Databases* (on page 10)" topic.
   a. Complete Steps 7-8, but do *not* perform Step 9 (cleanup old data). The old data is needed to determine how to upgrade the overrides.
6. Continue following the regular upgrade instructions until the test system is completely upgraded.
7. Generate updated parameter override files on the test system.
   a. Use CAD Query Tool to find parameter definition and default data changes.
      i. For parameter lists, run the following commands in CAD Query Tool:

      New and obsolete lists - `select SetName, ListName, AddedVersion, case Status when 0 then 'Obsolete' else 'New'`

`end Status from ParameterList where Status in (0,2) order by SetName, ListName`

New and obsolete parameter definitions - `select SetName, ListName, ParameterName, AddedVersion, case Status when 0 then 'Obsolete' else 'New' end Status from ParameterListParameterDefinition where Status in (0,2) order by SetName, ListName, ParameterName`

(Major Release Upgrades Only) New and obsolete parameter values - `select SetName, ListName, ParameterName, ModifiedVersion, case Status when 0 then 'Obsolete' else 'New' end Status, ParameterValue from ParameterListParameterValue where Status in (0,2) and ConfigurationName = 'default' order by SetName, ListName, ParameterName`

ii. For parameter tables, run the following commands in CAD Query Tool:

New and obsolete tables - `select SetName, TableName, AddedVersion, case Status when 0 then 'Obsolete' else 'New' end Status from ParameterTable where Status in (0,2) order by SetName, TableName`

New and obsolete columns - `select SetName, TableName, ColumnName, AddedVersion, case Status when 0 then 'Obsolete' else 'New' end Status from ParameterTableColumn where Status in (0,2) order by SetName, TableName, ColumnName`

(Major Release Upgrades Only) New and obsolete rows - `select SetName,TableName, RowNumber, ModifiedVersion, case Status when 0 then 'Obsolete' else 'New' end Status from ParameterTableRow where Status in (0,2) and ConfigurationName = 'default' order by SetName, TableName, RowNumber`

b. Update or add overrides as necessary in Administrator.

<u>Parameter Lists</u>

i. **New lists** - A new parameter list can be added in any release. Check if the default values for the parameters are sufficient.

ii. **Obsolete lists** - An obsolete list should no longer be used. It is removed in the next major release.

iii. **New parameters** - New parameters can be added to an existing list in any release. Check if the default values are sufficient.

iv. **Obsolete parameters** - Obsolete parameters should no longer be used. They are removed in the next major release.

v. **(Major Release Upgrades Only) New and obsolete values** - Default values can be changed for existing parameters in a major release. The new value displays with a status of *new* while the old value displays with a status of *obsolete*. Check if an override needs to be modified or created.

Parameter Tables

vi. **New tables** - A new parameter table can be added in any release. Check if the default rows are sufficient.

vii. **Obsolete tables** - An obsolete table should no longer be used. It is removed in the next major release.

viii. **New table columns** - New columns can be added to an existing table in any release. Check if override rows need to be modified or created.

ix. **Obsolete table columns** - Obsolete columns should no longer be used.

x. **(Major Release Upgrades Only) New and obsolete rows** - New or modified (new values) rows have a status of *new*. Deleted or modified (old values) rows have a status of *obsolete*. Check if override rows need to be modified or created.

Configuration Files

xi. Configuration files can be added, modified, or deleted in any release. A status value is added for configuration files in the next major release.

c. Unload the parameter overrides by running the following OcUtil command in PowerShell:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil"
UnloadCfgParam -d "$env:CustomParameters" -c allOverrides
```

d. Copy the unloaded files to a location that is accessible by the live system.

8. **(Major Release Upgrades Only)** Generate updated command line override files on the test system.

a. Use CAD Query Tool to find command line data changes.

New and obsolete rows -

```
select CommandKey, ModifiedVersion, case Status when 0 then
'Obsolete' else 'New' end Status from CommandKey where Status
in (0,2) order by CommandKey
```

```
select CommandKey, KeyIn, ModifiedVersion, case Status when
0 then 'Obsolete' else 'New' end Status from CommandLineKeyIn
where ConfigurationName = 'default' and Status in (0,2) order
by CommandKey, KeyIn
```

```
select CommandKey, KeyIn, ArgumentSwitch, ModifiedVersion,
case Status when 0 then 'Obsolete' else 'New' end Status from
CommandLineKeyInArgument where ConfigurationName =
'default' and Status in (0,2) order by CommandKey, KeyIn,
ArgumentSwitch
```

b. Update or add overrides as necessary in Administrator.

c. Unload the command line overrides by running the following OcUtil command in PowerShell:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil"
UnloadCommandLines -d "$env:CustomCommandLines" -c
allOverrides
```

d. Copy the unloaded files to a location that is accessible by the live system.

e. Exit upgrade mode by running the following OcUtil command in PowerShell:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil"
ZeroDowntimeUpgrade --complete
```

9. Generate updated string localizations and overrides on the test system.

   a. Review string definition changes in Administrator.

      i. **New strings** - New string keys can be added in any release. New string keys always have the default value of *US English*. Translations for other locales may not be immediately available. Review the new string keys to determine if additional localizations or overrides are necessary.

      ii. **Obsolete strings** - Obsolete string keys should no longer be used. They are removed in the next major release. String key definitions should not change after they are defined; however, if a substantial change is required, a new string key is defined, and the existing key is marked as *obsolete*.

   b. Update or add localizations and overrides, as necessary.

c. Unload the string localizations and overrides by running the following PowerShell command:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil"
UnloadStringValues -d "$env:CustomStrings"> -c allOverrides
```

d. Copy the unloaded files to a location that is accessible by the live system.

10. Generate files for custom dialog box data on a test system. The test system must have the same custom dialog boxes as the production (live) system.

a. *Modify custom dialog box data to work with updated default dialog boxes* (see "*Updating Custom Dialog Boxes after an Upgrade*" on page 45).

b. Unload the data in the CustomDialog database table by running the following CadBulkLoaderConsole command:

```
&
"$env:OnCallSystemRoot\cadbulkloader\CadBulkLoaderConsole
" /dbUser "$env:DbOwner" /dbUserPassword
"$env:DbOwnerPassword" /unloadTable CustomDialog /directory
"$env:CustomDialogs"
```

c. Copy the unloaded files to a location that is accessible by the live system.

Upgrade Your Live System

The zero downtime upgrade process can take some time. Intermediate points in the upgrade of the live system where it is possible to pause before proceeding are listed below.

- **Point 1 - Backups complete**

  Do not wait too long before proceeding. The longer you wait, the more data to be lost if you have to restore these from database backups.

- **Point 2 - First part of the database upgrade complete**

  At this point, the non-breaking database changes have been made, but no software has been updated on the application and web servers. The parameter list, the parameter table, command line, string, and custom dialog box data have also been updated. In the case of a major release, upgrade the parameter list, parameter table, and command line data are marked as old or new to prevent the software from seeing new, potentially incompatible data changes. If necessary, during a major release, it is possible to rollback just the parameter list, parameter table, and command line changes by using OcUtil to cancel the ZeroDowntimeUpgrade

mode. Do not do this after the software upgrade has been started because the new software depends on the new parameter list and parameter table data.

- **Point 3 - First half of application and web servers upgraded**

  At this point, half of the applications and web servers have been upgraded. The non-upgraded application servers have also been shut down. Existing clients are still running the old front end version of <aProductName>. These can interact with either the old or upgraded web servers because of internal component versioning that accounts for any changes. It is possible that newly-started clients or refreshed clients may load the new front end version of <aProductName>. Clients with the new front end version of <aProductName> only interacts with the upgraded web and application servers.

- **Point 4 - Second half of application and web servers upgraded**

  At this point, all of the application and web servers have been upgraded. Many, if not all, clients are still using the old front end version of <aProductName>. The clients are able to interact with all of the servers but are still dependent on the versioning mechanism to maintain compatibility.

- **Point 5 - Second part of database upgrade complete**

  The database upgrade is now complete. Old clients are still present.

- **Point 6 - Clients upgraded**

  All of the clients have now been upgraded to the new front end version of <aProductName>. The clients and servers are now using the upgraded version of <aProductName> so the versioning mechanism is no longer being used.

- **Point 7 - Cleanup completed**

  Now the cleanup of old parameter data and schema objects has been completed. The feature flags have also been updated to enable new functionality that is dependent upon them. The <aProductName> system has been completely upgraded at this point. However, in some cases, there may be additional software upgrades that should be made to update the database management system or RabbitMQ.

- **Point 8 - Live Database Management System upgraded**

  At this point, the Database Management System has been upgraded on all of the live database servers.

- **Point 9 - Archive Database Management System upgraded**

At this point, the Database Management System has been upgraded on all of the archive database servers. The only potential remaining work is to upgrade RabbitMQ.

The following procedures feature command examples for upgrading your database using PowerShell. However, if you are using Windows Authentication and a service account for upgrading your database, the command examples would instead look like the following:

```
-DbConfigName <DatabaseConfigName> -DbOwner "\"
-ArchiveDbConfigName <ArchiveDBConfigName> -ArchiveDbOwner "\"
```

Password parameters, such as ArchiveDbOwnerPassword, are not necessary to include when Windows Authentication is being used.

To perform a zero downtime upgrade on a live system:

In OnCall Dispatch systems running version 2303 or later, ISM is used to run the websites instead of IIS. It is necessary to stop 2209 websites in IIS, but the 2303 websites are started in ISM. Add-on products may or may not still use IIS. Hexagon recommends reviewing the documentation for each add-on for more information.

1. Create backups of the test system and the production (live) system.

   These backups can be done in parallel with one another.

   a. Make a backup of your live and archive databases before upgrading.

   b. Backup any changed or custom configuration files on all application and web servers. These may include the following:

      - ISM configuration files located in the *\CAD\ISM* folder
      - The *appsettings.json* files located in the application folders
      - The *log4net.config* files located in the application folders

2. Begin the upgrade for the databases.

   The live and archive databases can be upgraded in parallel with one another. However, the DBArchive service may experience errors during this process. These errors are typically resolved as you perform the upgrade process.

   a. Open a PowerShell window using the **Run as administrator** option on a computer that has access to the live and archive databases. Do not use an application or web server.

   b. From the PowerShell window, define the following environment variables using the *SetZDTUpgradeVariables.ps1* file:

- **OnCallSystemRoot** - This is where <aProductName> is or can be installed.
- **OnCallDeliveryRoot** - This is wherever your delivery media is installed.
- **OldVersion** - This is the current <aProductName> version.
- **NewVersion** - This is the new <aProductName> version.
- **DbConfigName** - This is the database configuration name for the live system.
- **DbOwner** - This is the database owner for the live system.
- **DbOwnerPassword** - This is the database owner password for the live system.
- **ArchiveDbConfigName** - This is the database configuration name for the archive system.

  If there is no archive database, leave this variable empty.
- **ArchiveDbOwner** - This is the database owner for the archive system.

  If there is no archive database, leave this variable empty.
- **ArchiveDbOwnerPassword** - This is the database owner password for the archive system.

  If there is no archive database, leave this variable empty.
- **StringLocales** - This is a list of one or more string locales, separated by commas.
- **CustomParameters** - This is the location of the parameter overrides that were generated on the test system.

  If there are no parameter overrides, leave this variable empty.
- **CustomCommandLines** - This is the location of the command line overrides that were generated on the test system.

  If there are no command line overrides, leave this variable empty.
- **CustomStrings** - This is the location of the string overrides and localizations that were generated on the test system.

  If there are no string overrides or localizations, leave this variable empty.
- **CustomDialogs** - This is the location of the custom dialog boxes that were generated on the test system.

If there are no custom dialog boxes, leave this variable empty.

- **CustomErrorRows** - This is the folder where CAD Bulk Loader places rows that could not be loaded.

  If there are no custom dialog boxes, leave this variable empty.

  An example of a full environment with defined variables looks like the following:

```
& "C:\delivery\IPS1122 - HxGN OnCall Dispatch -
Advantage\Advantage\SetZDTUpgradeVariables.ps1"
-OnCallSystemRoot "C:\Program Files\Intergraph"
-OnCallDeliveryRoot "C:\delivery\IPS1122 - HxGN OnCall
Dispatch - Advantage\Advantage" -OldVersion 10230300
-NewVersion 10230900 -DbConfigName live_db_config
-DbOwner live_db_config_owner -DbOwnerPassword
"password" -ArchiveDbConfigName archive_db_config
-ArchiveDbOwner archive_db_config_owner
-ArchiveDbOwnerPassword "password" -StringLocales
"en-us" -CustomParameters "c:\parameters"
-CustomCommandLines "c:\commandlines" -CustomStrings
"c:\strings" -CustomDialogs "c:\dialogs"
-CustomErrorRows "c:\error_rows"
```

c. Close any administration tools. Examples of administration tools include ISM Administrator, CAD Database Utility, CAD Query Tool, and CAD Bulk Loader.

d. Update the **Admin Tools** using the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
-AdminTools
```

e. Define the CAD and Database configurations in the *cad-global.json.config* file.

- These configurations are needed to let the tools connect to the database.
- The CAD configuration must be the default.
- The *cad-global.json.config* file can be copied from one of the existing servers.

f. Start upgrade mode using the following PowerShell command:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil"
ZeroDowntimeUpgrade --start --oldVersion "$env:OldVersion"
--newVersion "$env:NewVersion"
```

For a minor release or patch, the old version and the new version are the same. For example, in a minor upgrade from 10.2309.0 to 10.2309.1, set the old version and new versions to `10230900`.

g. **(Major Release Upgrades Only)** Make the non-breaking schema changes to the *live* database by running the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteUpgrade -BeginVersion "$env:OldVersion"
-DbConfigName "$env:DbConfigName" -User "$env:DbOwner"
-Password "$env:DbOwnerPassword" -EndVersion
"$env:NewVersion" -NonBreakingChangesOnly -Interactive
```

- You are prompted for values during the upgrade.
- This process can take some time.

h. **(Minor Release and Patch Upgrades Only)** To make non-breaking changes to your live database schema, run the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteMisc -SchemaFile
"$env:OnCallSystemRoot\sql\upgrade\$($env:NewVersion)fixe
s.xml" -DbConfigName "$env:DbConfigName" -User
"$env:DbOwner" -Password "$env:DbOwnerPassword"
-Interactive
```

You may be prompted for values during the execution of the file.

i. **(Major Release Upgrades Only)** Make the non-breaking schema changes to the *archive* database by running the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteUpgrade -BeginVersion "$env:OldVersion"
-DbConfigName "$env:ArchiveDbConfigName" -User
"$env:ArchiveDbOwner" -Password
"$env:ArchiveDbOwnerPassword" -EndVersion
"$env:NewVersion" -NonBreakingChangesOnly -Interactive
```

- You are prompted for values during the upgrade.
- This process can take a significant amount of time.

j. **(Minor Release and Patch Upgrades Only)** To make non-breaking changes to your *archive* database schema, run the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteMisc -SchemaFile
"$env:OnCallSystemRoot\sql\upgrade\$($env:NewVersion)fixe
s.xml" -DbConfigName "$env:ArchiveDbConfigName" -User
"$env:ArchiveDbOwner" -Password
"$env:ArchiveDbOwnerPassword" -Interactive
```

You may be prompted for values during the execution of the file.

k. Execute the *setup_archive_database.xml* file on the archive database by running the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteMisc -SchemaFile
"$env:OnCallSystemRoot\sql\setup_archive_database.xml"
-DbConfigName "$env:ArchiveDbConfigName" -User
"$env:ArchiveDbOwner" -Password
"$env:ArchiveDbOwnerPassword"
```

l. Update parameters, command lines, localized strings, custom dialog boxes, and other configuration data by running the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\ConfigurationDataUpgrade.ps1"
-OnCallSystemRoot  "$env:OnCallSystemRoot" -dbOwner
"$env:DbOwner" -dbOwnerPassword "$env:DbOwnerPassword"
-StringLocales "$env:StringLocales" -OverridesCfgParamPath
"$env:CustomParameters" -OverridesCommandLinesPath
```

```
"$env:CustomCommandLines" -OverridesStringDefinitions
"$env:CustomStrings" -CustomDialogs "$env:CustomDialogs"
-CustomErrorRows "$env:CustomErrorRows"
```

3. Update server software.
   a. In the load balancer, disable the first half of the OnCall Dispatch Websites to be upgraded.
   b. Determine which application and web servers to be upgraded in the first pass.

      In the same pass, the servers can be updated in parallel with one another.
   c. For each server, perform the following, applicable steps:
      i. Open a PowerShell window using the **Run as administrator** option.

         Do this for the application and web servers.
      ii. From the PowerShell window, define the following environment variables:

         Do this for the application and web servers.

- **OnCallSystemRoot**
  - This variable is typically *C:\Program Files\Intergraph*.
  - For example, `$env:OnCallSystemRoot = "C:\Program Files\Intergraph"`
- **OnCallDeliveryRoot**
  - This variable is wherever your delivery media is installed.
  - For example, `$env:OnCallDeliveryRoot = "Z:\delivery\IPS1122 - HxGN OnCall Dispatch - Advantage\Advantage"`

      i. Stop the ISM service (if it has not already been stopped) by running the following PowerShell command.

```
Stop-Service ism
```

- Do this for the application and web servers,
- If you have any-on products, you need to stop the websites for them, as well.

      i. Close any administration tools. Examples of administration tools include ISM Administrator, CAD Database Utility, CAD Query Tool, and CAD Bulk Loader.

Do this for the application and web servers.

ii. Update the **Admin Tools** packages by running the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
-AdminTools
```

- Do this for the application and web servers.
- This process can take some time.

iii. Update the services that were taken down by running the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
-Services
```

- Do this for *only* the application servers.
- This process can take some time.

iv. Update the interfaces that were taken down by running the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
-Interfaces
```

- Do this for *only* the application servers.
- This process can take some time.

v. Update the web applications that were taken down by running the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\UpgradeOnCallPackages.ps1"
-WebApps
```

Do this for *only* the web servers.

vi. Update the ISM configuration files.

- Do this for the application and web servers.
- This step overwrites any customized files.

On a primary server, run the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\OrganizeIsmConfigs.ps1"
-InstallDirectory $env:OnCallSystemRoot
```

On a secondary server, run the following PowerShell command:

```
& "$env:OnCallDeliveryRoot\OrganizeIsmConfigs.ps1"
-InstallDirectory $env:OnCallSystemRoot -Secondary
```

vii. Merge any ISM configuration file customizations back into the ISM configuration files by comparing the new files against the backups that were made earlier and updating the new files, as necessary.

Do this for the application and web servers.

viii. On the web servers, see the "Configure Your Web Servers" topic and perform the necessary steps.

This section includes steps on setting up the new global *appsettings.json* file.

ix. On the servers running RestfulCad and CadNotificationService, Hexagon recommends checking the **IssuerOverride** value in the *appsettings.json* files. See the "Multiple Instances and Tokens" topic in the *HxGN OnCall Dispatch* | *RestAPI Installation Guide* for more information.

x. On the web servers, expose port 44343 on the server firewall.

xi. Ensure the load balancer is accessing the upgraded web servers on port 44343.

xii. Start the ISM service by running the following PowerShell command:

```
Start-Service ism
```

Do this for the application and web servers.

xiii. Run **ISM Administrator** to monitor the services as they restart.

xiv. Wait until all of the services are running before proceeding.

xv. You may experience a heavy CPU load during this process. Run **Task Manager** to verify the CPU usage and wait until it is back to a normal level before proceeding.

xvi. If you have any add-on products, you need to start the websites for them, as well.

d. Stop the ISM service on all the application servers that have not been upgraded by running the following PowerShell command:

```
Stop-Service ism
```

Do this for *only* the application servers. If the website is also running on an application server, then do not shut down ISM. Instead, you need to stop the application services while leaving the following web services still running:

- **OnCall.Identity**
- **OnCall.Proxy**
- **OnCall.RealTime.WebApi**
- **OnCall.WebApi**
- **OnCall.Website**

e. In the load balancer, re-enable the websites that were upgraded.

f. In the load balancer, change the health check URL to the following endpoint:

```
/oncall/healthcheck
```

g. Wait for the system to stabilize before proceeding. This could take a few minutes.

h. In the load balancer, disable the remaining half of the websites that need to be upgraded.

i. Upgrade the remaining servers by repeating Steps 3c-3g.

4. Complete the database upgrade.

The live and archive databases can be upgraded in parallel with one another. However, the DBArchive service may experience errors during this process. These errors are typically resolved as you perform the upgrade process.

a. Continue on the computer where the database upgrade was started.

b. **(Major Release Upgrades Only)** Make remaining schema changes to the *live* database by running the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteUpgrade -BeginVersion "$env:OldVersion"
-DbConfigName "$env:DbConfigName" -User "$env:DbOwner"
-Password "$env:DbOwnerPassword" -EndVersion
"$env:NewVersion" -StartWithBreakingChanges -Interactive
```

c. Verify schema changes are consistent with the definition file for the *live* database by running the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteDefinition -SchemaFile
"$env:OnCallSystemRoot\sql\db_schema_both.xml"
-DbConfigName "$env:DbConfigName" -User "$env:DbOwner"
-Password "$env:DbOwnerPassword" -Interactive
```

d. **(Major Release Upgrades Only)** Make remaining schema changes to the *archive* database by running the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteUpgrade -BeginVersion "$env:OldVersion"
-DbConfigName "$env:ArchiveDbConfigName" -User
"$env:ArchiveDbOwner" -Password
"$env:ArchiveDbOwnerPassword" -EndVersion
"$env:NewVersion" -StartWithBreakingChanges -Interactive
```

e. Verify schema changes are consistent with the definition file for the *archive* database by running the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteDefinition -SchemaFile
"$env:OnCallSystemRoot\sql\db_schema_both.xml"
-DbConfigName "$env:ArchiveDbConfigName " -User
"$env:ArchiveDbOwner" -Password
"$env:ArchiveDbOwnerPassword" -Interactive
```

f. Execute the *setup_archive_database.xml* file on the *archive* database by running the following PowerShell command:

```
&
"$env:OnCallSystemRoot\cad_db_utility\CadDbUtilityConsole
" -ExecuteMisc -SchemaFile
"$env:OnCallSystemRoot\sql\setup_archive_database.xml"
-DbConfigName "$env:ArchiveDbConfigName" -User
"$env:ArchiveDbOwner" -Password
"$env:ArchiveDbOwnerPassword"
```

5. Migrate all of the web and application clients to the new (upgraded) version.

- The web and application clients can be upgraded in parallel with one another.
- This process can take some time as your site runs both old and new clients.

a. Use Administrator to send out a notification the upgrade has completed.
   i. Navigate to the **Administration > System Notification Manager** page.
   ii. Click **Send Upgrade Complete Notification**.

b. Refresh each web client when they are not busy.

c. Logout of application clients and restart them when they are not busy.
   i. During this step, you may be prompted to install a newer version of the application client (if one is available). Accept the new application client before proceeding.

6. Perform cleanup steps.

   a. Continue on the computer where the database upgrade was completed.

   b. Exit upgrade mode by running the following OcUtil command in PowerShell:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil"
ZeroDowntimeUpgrade --complete
```

   c. Update feature flags by running the following OcUtil command in PowerShell:

```
& "$env:OnCallSystemRoot\OnCall\OcUtil\OcUtil"
FeatureFlags -f
"$env:OnCallSystemRoot\OnCall\OcUtil\ConfigurationData\Fe
atureFlags\feature_flags.config.json " -v -d
```

   d. **(Major Release Upgrades Only)** Clean up the database by running the following PowerShell command.

      Archive parameters are only needed if there is an archive database.

```
& "$env:OnCallDeliveryRoot\DatabaseUpgradeCleanup"
-OnCallSystemRoot  "$env:OnCallSystemRoot" -LiveDbConfig
"$env:DbConfigName" -LiveDbOwner  "$env:DbOwner"
-LiveDbPassword "$env:DbOwnerPassword" -ArchiveDbConfig
"$env:ArchiveDbConfigName" -ArchiveDbOwner
"$env:ArchiveDbOwner" -ArchiveDbPassword
"$env:ArchiveDbOwnerPassword"
```

7. If necessary, perform a rolling upgrade for the Database Management System (DBMS) on the *live* database servers.

8. If necessary, perform a rolling upgrade for the DBMS on the *archive* database servers.
9. Upgrade RabbitMQ.

   If there is an update to the RabbitMQ version that is delivered, use the instructions in the "Automatic Installation and Configuration" topic in the *RabbitMQ Installation and Configuration Guide*.

Customizable File Changes

If you have customized files for your site, you should evaluate the customized files to ensure the customizations are still appropriate; if so, you must merge the updated version of the base files with the customized version of the files to retain your customizations.

- If you have customized XSL templates for your site, you must use Xalt to merge the updated version of the templates with the customized version.
- If you lost any customized templates during your upgrade, you need to re-create them. If the application supports reading the customized templates from configuration files, Hexagon recommends you upload your customized templates to the database using the **Configuration Files** page in Administrator. By uploading your customized templates files to the database, they are not lost with future upgrades.

Updating Custom Dialog Boxes after an Upgrade

When updating your version of HxGN OnCall Dispatch software, custom changes to dialog boxes should be automatically merged with the latest update, that is, your changes should be preserved. If changes to dialog boxes do not display after an update or restart of the DialogModules service in ISM, you most likely need to manually re-apply your changes to the default dialog box. It is best to remove your updated dialog box and create the dialog box with custom changes by saving the dialog box to a specific configuration again.

For information on using UI Editor to modify or create dialog boxes, see the "UI Editor" topic in the *HxGN OnCall Dispatch | Administrator Help*.

Appendix

Use Chocolatey, NuGet, and Scripts to Install and Uninstall Hexagon Software

Most of the Hexagon software, such as HxGN OnCall Dispatch and the system administration tools, are installed using Windows batch files. The batch files call a PowerShell script called S*etup.ps1*, which uses Chocolatey and NuGet to install the Hexagon software. If this is the first time you are running this script, you are prompted to accept that you trust Hexagon as a publisher. Select `Always run` on this prompt.

You can run the Windows batch files to install and uninstall Hexagon software, but you may be interested in how the installation process works.

NuGet is a package management tool from Outercurve Foundation and is an open source under the Apache License 2.0. A NuGet package file is a ZIP file with a known structure. You can temporarily change the extension of a NUPKG file to ZIP to view the contents. If you do this, make sure to change back to a NUPKG file before running any installation files.

Chocolatey is a tool for automating software installations. Chocolatey relies on NuGet and PowerShell. Chocolatey supports delivering software from internal or external package feeds, but HxGN OnCall Dispatch uses NUPKG files delivered in the *\packages* folder of your installation media. You can interact with Chocolatey using the command line tool chocolatey (choco). You can use choco to view the list of software installed locally by Chocolatey by running `choco list -lo`. You can install software using `choco install` and uninstall software using `choco uninstall`. To get more information about a command, type the command followed by /?. For example, to see the Help for the choco install, run `choco install /?`.

The Hexagon installation scripts install Chocolatey if it is not already installed. If you experience issues with the automatic installation of Chocolatey, you can manually install Chocolatey following the instructions *here* (*https://chocolately.org/install*). Make sure to run the Chocolatey installation using a prompt that you open with the **Run as administrator** option. By default, Chocolatey is installed to *\ProgramData\chocolatey* on your system drive. Note that *\ProgramData* is a hidden folder, so it is not shown by default in Windows File Explorer. To view the *\ProgramData folder*, click the **View** menu in File Explorer and check **Hidden items**.

Configure Firewalls

The HxGN OnCall Dispatch system requires multiple firewall configurations in order for products to communicate properly. The following table demonstrates some configuration information, as well as default ports and protocols, that you need to follow when setting up your network.

In addition to configuring the firewall for HxGN OnCall Dispatch applications, you must configure your application servers to support remote access.

Product	Protocol	Listener Ports	Destination Ports	Destination Product	Configuration Instructions	Database Connection	RabbitMQ Connection
ISM				ISM instance on peered server	ISM now uses the RabbitMQ connection that listener uses.	No	Yes
ISMAdmin				ISM service	ISMAdmin no longer uses named pipes. All communicatio n is done over RabbitMQ.	No	Yes
CAD Connect for EdgeFrontier		135/UDP 389/TCP 1801/TC P 1081/UD P 2101/UD P 2103/UD P 2105/UD P 3527/UD P		IFCADDirect	These are the port numbers that must be open for MSMQ to run.	No	

Product	Protocol	Listener Ports	Destination Ports	Destination Product	Configuration Instructions	Database Connection	RabbitMQ Connection
Mobile Responder	HTTPS					No	No
CAD Notification Service	HTTP or HTTPS	443/TCP or 8081/TCP		Custom application The custom application informs CNS of the callback URL, which includes the port to use.	Default HTTPS request	Yes	Yes
CAD Database Utility						Yes	No
CAD Query Tool						Yes	No
Change Publisher Service						Yes	Yes
Database Copy						Yes	No

Product	Protocol	Listener Ports	Destination Ports	Destination Product	Configuration Instructions	Database Connection	RabbitMQ Connection
Deccan Interface	HTTP or HTTPS	443/TCP or 8088/TCP	443/TCP	Custom application When using SSL, the site administrator determines the SSL port to be used by each custom application, which all default to port 443 when SSL. Deccan uses 8088 by default.	Default HTTPS request	Yes	Yes
LOI Service						Yes	Yes
OnCall.Proxy	HTTPS	44343/TCP	44343/TCP	*OnCall.Proxy.exe*	Default HTTPS request	No	Yes

Product	Protocol	Listener Ports	Destination Ports	Destination Product	Configuration Instructions	Database Connection	RabbitMQ Connection
RestAPI	HTTP or HTTPS	443/TCP or 8080/TCP	443/TCP	Custom application When using SSL, the site administrator determines the SSL port to be used by each custom application. RestAPI uses port 8080 by default.	Default HTTPS request	Yes	Yes
Tracker	UDP	21000/UDP	21000/UDP	When the Transport parameter is set to UDP in the HxGN OnCall Dispatch | Tracker GeneralDetails parameter list.	Set the UdpReadPort parameter to the value wanted from 21000 through 21099 (default=21000).	No	Yes

Product	Protocol	Listener Ports	Destination Ports	Destination Product	Configuration Instructions	Database Connection	RabbitMQ Connection
Tracker	UDP	21099/UDP	21099/UDP	When the Transport parameter is set to UDP in the HxGN OnCall Dispatch | Tracker GeneralDetails parameter list and PassiveMode is set to 0.	Set the UdpSendPort parameter to the value wanted from 21000 through 21099 (default=21099).	No	Yes
Tracker	CAD	RabbitMQ	RabbitMQ	When the Transport parameter is set to CAD in the HxGN OnCall Dispatch | Tracker GeneralDetails parameter list.		No	Yes
Scheduler Service						Yes	Yes

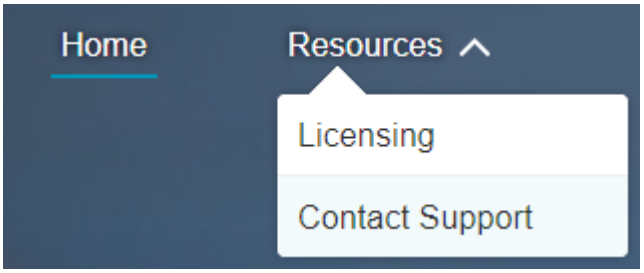
Technical Support

Hexagon's Safety, Infrastructure & Geospatial division provides several ways to contact Support, including online case submission and phone support. For general Hexagon information, call 800-345-4856 (US). For phone support, see the *Contact Support page https://supportsi.hexagon.com/help/s/contactsupport?language=en_US*.

Submit a Case Online

Hexagon provides an online method of requesting support 24/7.

1. Go to the *Support page https://hexagon.com/support-success/safety-infrastructure-geospatial*.
2. Click **Log in**.
3. Type your username and password, and click **Log in**.
4. From the **Resources** list, select **Contact Support**.



   The **Contact Support** page displays.
5. Click **Online case submission**.